Worst-Off or Worst Case?

The Structure of Lexical Difference Principles under Ignorance

Abstract

Under uncertainty about the state of the world and ignorance of probabilities, the lexical difference principle (LDP) takes several forms that rank policies differently. I show that these forms differ in two independent choices: what a rule keeps track of (which outcomes belong to the same person, or which occur in the same state), and where lexical priority goes first (to the worst-off, the person or position at the bottom of society, or to the worst case, the state or outcome at the bottom of the range of uncertainty). Representing policies as person-by-state matrices, I show that the non-basic specifications identified by De Coninck and Van De Putte (2023) fill three of the four combinations of the two choices, and that transposing the matrix yields the fourth, a new atomistic ex ante LDP. Each atomistic rule is its holistic counterpart evaluated under the least favorable rearrangement of the matrix. Rules that track persons violate one of De Coninck and Van De Putte's (2025) indifference conditions, and rules that track states violate the other. The ex ante/ex post distinction, as usually drawn, bundles the two choices, and Diamond's case tests both. Neither the standard original position nor Rawls's own reasoning settles either.

Keywords Difference principle · Leximin · Original position · Ex ante and ex post evaluation · Choice under ignorance · Duality

1 Introduction

The lexical difference principle (LDP) ranks alternatives by the welfare of the worst-off, then, among alternatives tied at that level, by the welfare of the second worst-off, and so on. Many take it to be the best formal statement of Rawls's priority to the worst-off (Section 2.1). Under uncertainty about the state of the world, however, the lowest outcome is still well defined, but what the principle should compare next is not.

Diamond's (1967) case, in the form used by De Coninck and Van De Putte (2023), shows how (Table 1). Two persons face three policies, and the outcome of each depends on which of two states occurs, and nothing is known about how likely the states are. Under policy a, person 1 receives 0 and person 2 receives 1 in both states. Under b, the state decides who receives 0. Under

c, both receive 0 in state 1 and both receive 1 in state 2. Many judge b fairer than a, since it does not let the loss always fall on the same person.

Table 1 Diamond's case (De Coninck & Van De Putte, 2023, Fig. 7; welfare of persons 1 and 2)

Alternative	State 1	State 2
a	(0, 1)	(0, 1)
b	(0, 1)	(1, 0)
c	(0, 0)	(1, 1)

Each of the lexical rules studied below continues differently after the lowest outcome, and each selects a different set of policies. Here I describe them informally; Section 3 defines them. Holistic rules compare whole units, a person's prospect or a state's distribution; atomistic rules compare rank by rank. The holistic ex ante rule (E) follows persons and compares their whole prospects, worst prospect first. Person 1's prospect under a is (0, 0), whereas every prospect under b and c is (0, 1), so E rejects a and is indifferent between b and c. The atomistic ex post rule (A) follows social positions rather than persons (a position is a rank within a state, whoever holds it): it compares how the bottom position fares across states, then the next position, and so on. The bottom position receives 0 in both states under a and b but 1 in state 2 under c, so A selects c. The holistic ex post rule (H) compares the distributions in the states, worst state first. Under c the worst state gives (0, 0), whereas under a and b every state gives (0, 1), so H selects a and b. Finally, the atomistic ex ante rule (P) first compares everyone's worst case, then everyone's second-worst case. Under a, person 2's worst case is 1; under b and c, everyone's worst case is 0. So P selects a, the policy in which the same person loses in every state. A fifth, basic rule (B), which pools all outcomes, cannot tell the three policies apart.

What separates these verdicts are two choices, stated informally here and precisely in Section 4. The first is the *information choice*: what a rule keeps track of. A rule that tracks persons, or retains person-linkage, knows which outcomes belong to the same person. For instance, it knows that the two 0s person 1 receives under a fall on a single life, but ignores which outcome occurs in which state. A rule that tracks states, or retains state-linkage, knows which outcomes occur together. For instance, it knows that the two 0s under c occur together in state 1, but ignores which outcome belongs to which person. E and P track persons, and so cannot tell b from c: under both, each person's prospect is (0, 1). A and H track states, and so cannot tell b from a: under both, each

state's distribution is (0, 1). B tracks neither, and so cannot tell any of the three apart.[1]

The second is the *priority choice*[2]: after the lowest outcome, whether the comparison continues with the worst-off or with the worst case. E and A give priority to the worst-off, the person or position at the bottom of society, and prefer c to a: under c, unlike under a, no person and no position receives 0 in every state. H and P give priority to the worst case, the state or outcome at the bottom of the range of uncertainty, and prefer a to c: under a, unlike under c, someone receives 1 even in the worst case. Under both a and c, person 1 is never better off than person 2, and state 1 is better for no one. Tracking persons or states makes no difference to the comparison between a and c, so this disagreement turns on priority alone (Section 4.4). Each of the four rules combines one answer to each choice (Table 2), and the basic rule makes neither choice.

Table 2 The two choices

	Tracks persons (ex ante)	**Tracks states (ex post)**
Priority to the worst-off	E: worst-off person	A: bottom position
Priority to the worst case	P: everyone's worst case	H: worst state

The rules usually contrasted as ex ante and ex post are E and H. Yet they differ in both choices: E tracks persons and gives priority to the worst-off, whereas H tracks states and gives priority to the worst case. The usual contrast therefore varies the two choices together (Section 4.3). Diamond's case, the standard test of ex ante fairness, likewise tests both choices at once (Section 4.4). The new rule P demonstrates this: although it tracks persons, it selects the policy in which the same person loses in every state. Tracking persons is therefore insufficient for ex ante fairness; priority must also go to the worst-off. Since the familiar uses of the terms bundle the two choices, I reserve them for the information choice in what follows, as in Table 2: ex ante rules track persons, and ex post rules track states (Section 2.2 discusses other readings).

This paper builds on De Coninck and Van De Putte (2023), who distinguish four specifications of the LDP under ignorance, B, E, A, and H, and prove that the standard original position supports

[1]The information choice is not forced: a rule can retain both kinds of linkage, for instance by combining two of these rules lexicographically (Section 4.4). Each of the five rules studied here, however, retains at most one, because it sorts outcomes within persons, within states, or all together before comparing them, and sorting forgets the other dimension's information.
[2]Priority here means lexical precedence, not the weighting of prioritarianism.

only the basic one. Their non-basic rules fill three cells of Table 2. I find the fourth by writing each policy as a person-by-state matrix and exchanging the roles of persons and states, that is, by transposing the matrix: E is the transpose of H, and B is its own transpose, but the transpose of A is none of the four. It is a new rule, P.

The two atomistic rules arise differently. A grows out of the language of the difference principle itself, which speaks of the worst-off, the second worst-off, and so on. For Rawls, these are positions rather than particular persons: "when principles mention persons … the reference is to representative persons holding the various social positions" (1999, p. 56). P comes into view through a formal method, here transposition. This is one reason why formalizing the structures of rules matters: intuition is powerful, but not comprehensive. A family of rules not closed under exchanging persons and states treats the two asymmetrically without saying so; once it is closed, any asymmetry between persons and states must be argued for rather than built into the options.

Section 2 reviews the background and related literature. Section 3 explains the setup, defines the five rules, and proves their distinctiveness and transposition structure. Section 4 proves the structure between holistic and atomistic rules. Section 5 relates the information choice to De Coninck and Van De Putte's (2025) two indifference conditions and applies this relation to the five rules: only the basic rule is supported, so the standard original position settles neither choice. Section 6 shows that Rawls's own reasoning settles neither choice, and argues that a Rawlsian who favors a particular lexical principle must therefore defend both on independent grounds. Section 7 concludes.

2 Background and related literature

2.1 The lexical difference principle

Read as a social ranking, the difference principle compares alternatives by the position of the worst-off alone. It therefore violates strong Pareto: whenever the worst-off fare equally under two alternatives, it ranks them as equally good, even if one is better for others and worse for no one (Sen, 1970). The standard repair is lexical: if the worst-off are tied, compare the second worst-off, then the third, and so on. Rawls states this lexical refinement himself (1999, p. 72). Under ignorance, ties at the minimum are common, and the specifications of the principle diverge only at the lexical level (Section 2.4). Parfit (2000, p. 120) and Van Parijs (2002, pp. 208–209) argue

that the original position itself supports the lexical principle: a party who does not know whether they will be among the worst-off has reason to care about the better-off positions as well, provided this does not make the worst-off worse off.

2.2 Ex ante and ex post

When outcomes depend on the state of the world, so that a person's welfare may differ from state to state, a policy is naturally represented as a matrix with persons as rows and states as columns (Ben-Porath et al., 1997; Diamond, 1967). Diamond's (1967) example, a critique of Harsanyi's (1955) utilitarianism, shows why this representation matters. Two policies can yield the same distribution in every state and differ only in who bears the loss: under one, the same person loses in every state; under the other, a fair coin decides who loses (policies a and b in Table 1). In this paper, under ignorance, the coin is replaced by the state: under b, person 1 loses in one state and person 2 in the other. Many judge the second fairer, yet Harsanyi's utilitarianism must treat the two as equivalent.

From Diamond's example grew the debate over ex ante and ex post evaluation. The contrast can be drawn in several ways and in several settings. Most of the literature uses a probabilistic setting, as in Diamond's coin flip (Mongin & Pivato, 2016); a second setting is ambiguity, where probabilities are imprecise (Rowe & Voorhoeve, 2018; Inoue & Miyagishima, 2022); a third is ignorance, which can be read as the limiting case of ambiguity. The contrast has rarely been drawn under ignorance (but see De Coninck & Van De Putte, 2023; Section 2.4), although decision under ignorance has a literature of its own (Luce & Raiffa, 1957, ch. 13; Maskin, 1979). To define the two terms under ignorance, I therefore borrow from the probabilistic literature. Two readings matter here.

On a procedural reading, ex ante approaches evaluate each person's prospect as a whole and then compare persons, whereas ex post approaches evaluate each state's distribution as a whole and then aggregate across states (Fleurbaey, 2010; Mongin & Pivato, 2021). The procedural reading fixes more than information. Comparing persons' whole prospects tracks persons and also gives priority to the worst-off, and comparing states' whole distributions tracks states and also gives priority to the worst case. The procedural reading therefore bundles the information choice and the priority choice, and the atomistic rules, in which the two choices diverge, have no stable place on it (Section 4.3). On an informational reading, which I adopt, an evaluation is ex ante if it tracks

persons, and ex post if it tracks states. This follows the axiomatic practice of distinguishing the two approaches by the Pareto principle they respect, since ex ante Pareto is stated in terms of persons' prospects and ex post Pareto in terms of states' distributions (cf. Mongin & Pivato, 2016). Intuitively, a rule that tracks persons looks at what each person faces before the state is known, and a rule that tracks states looks at what happens once it is. In Section 4.3, I argue that the informational reading is better.

Ex ante and ex post conflict in familiar cases, and several criteria combine them (Ben-Porath et al., 1997; Gajdos & Maurin, 2004; Mongin & Pivato, 2021). A related literature characterizes egalitarian and maximin criteria under severe uncertainty or cautious attitudes to risk (Fleurbaey & Zuber, 2017; Inoue & Miyagishima, 2022; Rowe & Voorhoeve, 2018). This work relies on probabilities or on additive or maximin criteria, rather than on lexical comparisons under ignorance.

2.3 Two kinds of ignorance

Two sources of uncertainty should be kept apart. *Identity uncertainty* concerns which position in society one will occupy; *world uncertainty* concerns external contingencies such as growth or recession. Rawls clearly excludes probabilistic reasoning about the first. His treatment of the second is less settled: the parties "have no basis for determining the probable nature of their society" (Rawls, 1999, p. 134), yet what they are to discount are "estimates of likelihoods not supported by a knowledge of particular facts" (Rawls, 1999, p. 150). Gustafsson (2018, n. 28) accordingly argues that objective probabilities of risks within society are admissible, since without them the parties could not assess how principles distribute risk. The rules studied below show that such an assessment does not require probabilities.

De Coninck and Van De Putte (2023) model world uncertainty as ignorance, with finitely many states and no probabilities, and introduce identity uncertainty separately through their original-position transformation (Section 5).[3] I follow them, both because their taxonomy is defined in this setting and because Rawls's argument against Harsanyi takes the veil to exclude any basis for assigning probabilities to states. This assumption is crucial: if the states in Table 1 had probabilities of 1 and 99 percent, states would carry weights that persons do not, and the symmetry between

[3] Unlike maximin, lexical rules are sensitive to duplicating a state, which Luce and Raiffa's (1957, p. 295) Axiom 11 for complete ignorance rules out; Gustafsson (2022) shows that every plausible rule for decisions under ignorance has some such sensitivity. Like De Coninck and Van De Putte, I take the set of states as given.

rows and columns would break.

Rawls states the difference principle in terms of the expectations of representative persons (Rawls, 1999, p. 56), which he defines as the index of primary social goods "which a representative individual can look forward to" (Rawls, 1999, p. 79).[4] A representative person stands for a social position rather than a particular individual.

2.4 Lexical difference principles under ignorance

De Coninck and Van De Putte (2023) show that three natural specifications of the simple, maximin difference principle coincide under ignorance (their Observation 3), since each ranks alternatives by the lowest outcome: the basic one, the ex ante one with a pessimistic measure of a person's welfare, and the ex post one. Divergence appears only at the lexical level, where they distinguish the four specifications B, E, A, and H illustrated in Section 1 and defined in Section 3.2, and show that only B, which cannot distinguish the alternatives in Diamond's case, can be derived from an original-position argument (in the sense explained in Section 2.5).

Two features of their treatment are relevant here. First, for the simple principle they consider two measures of a person's ex ante welfare: a pessimistic measure, the person's worst outcome, and a lexical measure, the whole ordered prospect (De Coninck & Van De Putte, 2023, Section 3.1). Only the lexical measure of ex ante welfare is carried into their specifications of the LDP, yielding the holistic ex ante rule. Second, their labels are asymmetric: the ex post rules are qualified as atomistic or holistic, but the ex ante rule is not. Together, these point to a rule missing from their specifications.

2.5 Why the original position cannot adjudicate

De Coninck and Van De Putte (2025) extend the diagnosis of their 2023 paper. An original-position argument, in their sense, derives a social ranking from the choice of a single party who does not know which person they will be: the party treats this as a further source of ignorance, alongside the state of the world, and chooses by an individual decision rule that is impartial among persons and among states. They show that a rule can be supported by an original-position argument if and only if it satisfies two indifference conditions, Indifference to Intra-State Distribution of Payoffs to Persons (IISD) and Indifference to Intra-Person Distribution of Payoffs at States (IIPD)

[4] Despite the word, these expectations are not probability-weighted averages over states.

(Section 5), which together erase the information that some outcomes belong to the same person and some to the same state. They also call for a model with several dimensions of uncertainty. The contrast between person-linkage and state-linkage used below is thus already present in their conditions; what is new here is the second, independent choice of priority (Corollary 3).

2.6 Related work and the gap

Mongin and Pivato (2015) also treat persons and states symmetrically in a matrix framework, but to derive additively separable rankings rather than the lexical ones under ignorance. Maskin (1979) already reads each state as a person, but to characterize leximin for a single decision-maker, rather than in a person-by-state framework. Statewise evaluation has been characterized for maximin under severe uncertainty (Inoue & Miyagishima, 2022); lexicographic decision rules are well studied (Fishburn, 1974); and leximin is well understood as an individual criterion under ignorance (Barberà & Jackson, 1988). The row-sorting operation used below also appears in the philosophy of inequality over time: Lippert-Rasmussen's (2003) ranking-order segment account compares segments of different lives that hold the same rank within each life.

To my knowledge, no prior work separates the priority choice from the information choice in lexical difference principles, identifies the missing transpose dual in De Coninck and Van De Putte's taxonomy, or shows that each atomistic rule evaluates policies as its holistic counterpart would under the least favorable rearrangement.

3 The rules and the duality

3.1 Setup

This section fixes the notation. Following De Coninck and Van De Putte (2023), a choice scenario is a tuple $C = (N, \mathcal{A}, S, d)$, where $N = \{1, \dots, n\}$ is a finite set of persons, $\mathcal{A}$ is a finite set of alternatives (e.g., the policies in Table 1), $S = \{1, \dots, m\}$ is a finite set of states, and $d(i, a, s)$ is the welfare[5] of person i under alternative a in state s. There are no probabilities over states. For each alternative a, let X_a be the $n \times m$ matrix whose entry in row i and column s is $d(i, a, s)$. De Coninck and Van De Putte list outcomes state by state, with the persons' welfare levels nested

[5]Welfare need not be cardinal utility. Only comparisons of welfare levels matter, across persons and states: every rule below is invariant under any common strictly increasing transformation of *d*. Welfare can be read as Rawls's index of primary goods.

within each state. The matrix form treats persons and states symmetrically, which makes transposition available, and it lets every rule be applied to any matrix, including rearranged and transposed ones.

For a vector v, $\mathrm{sort}(v)$ is its rearrangement in nondecreasing order, and $v[k]$ is its k-th entry. The lexicographic order $\geq_L$ on vectors of equal length is defined by $v \geq_L w$ if $v = w$ or, at the first coordinate where they differ, the entry of v is larger. On tuples of vectors, $\geq_L$ compares the vectors in order, each by $\geq_L$. For a list of vectors, $\mathrm{sort}_L(\cdot)$ arranges them from lowest to highest under $\geq_L$. For a matrix X, let $r_i(X) = \mathrm{sort}(\text{row } i \text{ of } X)$, person i's outcomes from worst to best, and $c_s(X) = \mathrm{sort}(\text{column } s \text{ of } X)$, the distribution in state s from worst to best.

I use De Coninck and Van De Putte's (2023, Fig. 6) example (Table 3) as a running example. For alternative c, person 1 receives 5 in state 1 and 3 in state 2, and person 2 receives 1 and 2. Person 1's sorted outcomes are therefore (3, 5) and person 2's are (1, 2); the sorted distributions are (1, 5) in state 1 and (2, 3) in state 2.

Table 3 De Coninck and Van De Putte's example (welfare of persons 1 and 2)

Alternative	State 1	State 2
a	(2, 1)	(3, 4)
b	(3, 3)	(1, 3)
c	(5, 1)	(3, 2)
d	(4, 4)	(1, 2)

Each rule F below assigns to every matrix X a *key* $F(X)$, the counterpart of what De Coninck and Van De Putte call the value of an alternative: the information relevant to its ranking, which need not be a single number (2023, p. 287, n. 17). A rule ranks a at least as high as b if and only if $F(X_a) \geq_L F(X_b)$. Each key discards some labels by sorting and fixes the order in which the remaining information is compared; the rules below differ in both respects. De Coninck and Van De Putte's social choice rules, however, are choice correspondences: they return a set of admissible alternatives, not a ranking. Each rule F therefore induces the correspondence

$$S_F(C) = \{a \in \mathcal{A} : F(X_a) \geq_L F(X_b) \text{ for all } b \in \mathcal{A}\}.$$

Following De Coninck and Van De Putte (2025), results about invariance and original-position support are stated for the choice sets S_F; results about keys and rankings, such as Observation 1,

are stated for F itself and imply the corresponding results for S_F.

3.2 Five lexical rules

This section states De Coninck and Van De Putte's (2023) four rules in matrix form and introduces the new one.[6]

Definition 1 (Basic, B). $B(X)$ is the sort of all $n \cdot m$ entries of X. This is leximin applied to the pooled list of person–state outcomes. In the running example, $B(X_c) = (1,2,3,5)$.

Definition 2 (Holistic ex ante, E). $E(X) = \mathrm{sort}_L(r_1(X), \dots, r_n(X))$. Alternatives are compared by the worst person's whole prospect first, then the second-worst person's, and so on. For c, the worst prospect is person 2's, (1, 2), so $E(X_c) = ((1,2),(3,5))$; for d, $E(X_d) = ((1,4),(2,4))$, so E ranks d above c.

Definition 3 (Holistic ex post, H). $H(X) = \mathrm{sort}_L(c_1(X), \dots, c_m(X))$. Alternatives are compared by the worst state's whole distribution first, then the second-worst state's, and so on. For c, state 1 is the worse state, since (1, 5) is lexicographically below (2, 3), so $H(X_c) = ((1,5),(2,3))$.

Definition 4 (Atomistic ex post, A). $A(X) = (\alpha_1, \dots, \alpha_n)$, where

$$\alpha_k = \mathrm{sort}(c_1(X)[k], \dots, c_m(X)[k]), \quad k = 1, \dots, n.$$

For each rank k, α_k records how the k-th worst social position fares across states. Alternatives are compared first by α_1, the bottom position, then by α_2, and so on. For c, the bottom position receives 1 in state 1 and 2 in state 2, so $\alpha_1 = (1,2)$ and $A(X_c) = ((1,2),(3,5))$. This coincides with $E(X_c)$ because person 2 is worse off in both states; Section 4 explains why.

The fifth rule is new.

Definition 5 (Atomistic ex ante, P). $P(X) = (\beta_1, \dots, \beta_m)$, where

$$\beta_k = \mathrm{sort}(r_1(X)[k], \dots, r_n(X)[k]), \quad k = 1, \dots, m.$$

For each k, β_k collects every person's k-th worst outcome. Alternatives are compared first

[6] In their notation, $B(X_a)$, $E(X_a)$, $A(X_a)$, and $H(X_a)$ are the ordered versions of the tuples they write as $d_b(a)$, $D_e(a)$, $D_w(a)$, and $D_s(a)$ (De Coninck & Van De Putte, 2023, Section 3.1 and Definitions 9–11), and $\geq_L$ combines their lexicographic ranking and lexicographic meta-ranking.

by β_1, the vector of all persons' worst-case outcomes, by leximin; if tied, by β_2, all persons' second-worst outcomes; and so on. For c, the persons' worst cases are 3 and 1, so $\beta_1 = (1,3)$ and $P(X_c) = ((1,3),(2,5))$.

The name atomistic ex ante follows De Coninck and Van De Putte's usage: A is atomistic because it compares rank by rank rather than unit by unit, and P does the same; Section 4.3 gives a more precise characterization. The first layer of P, β_1, collects De Coninck and Van De Putte's pessimistic measure of ex ante welfare for every person (Section 2.4), so P is a lexical extension of that measure. The letters B, E, H, and A follow their names; P recalls the pessimistic measure.

De Coninck and Van De Putte (2023, Section 4.1) require lexical specifications to satisfy two desiderata.[7] Lemmas 1 and 2 show that P meets both desiderata and strong Pareto, as the four rules do. Since the first entry of β_1 is the lowest outcome, P also refines the simple difference principle, so it is a lexical specification of that principle in the same sense as theirs.

Lemma 1 (Desiderata) (i) In a strictly egalitarian scenario, where all persons have the same welfare in each state, P agrees with individual leximin over states. (ii) In a basic scenario, with a single state, P agrees with leximin over persons.

Proof (i) If all rows of X are identical, every $r_i(X)$ equals the same vector r, so $\beta_k = (r[k], \dots, r[k])$ for each k, and comparing $(\beta_1, \dots, \beta_m)$ lexicographically is equivalent to comparing r lexicographically.

(ii) If $m = 1$, then $P(X) = (\beta_1)$, where β_1 is the sorted vector of the persons' outcomes. □

Lemma 2 (Strong Pareto) If X weakly exceeds Y in every entry and strictly exceeds it in at least one, then $P(X) >_L P(Y)$. The same holds for B, E, H, and A.

Proof Increasing entries of a vector weakly increases every entry of its sorted rearrangement. Hence $r_i(X) \geq r_i(Y)$ coordinatewise for every i, and therefore $\beta_k(X) \geq \beta_k(Y)$ coordinatewise for every k. Since the entries of X sum to more than those of Y, some $\beta_k(X) \neq \beta_k(Y)$. At the first

[7] In their words: "First, when applied to a strictly egalitarian scenario, the rules give the same recommendation as the leximin rule would give to any of the individuals. Second, when applied to a basic scenario, they give the same recommendation as the leximin rule would give, when applied to the original position transformation of that scenario" (De Coninck & Van De Putte, 2023, p. 290). In a basic scenario, with a single state, leximin applied to the original-position transformation coincides with leximin over persons, which is the form used in Lemma 1(ii).

such k, $\beta_k(X)$ weakly exceeds $\beta_k(Y)$ in every coordinate and differs from it, so $\beta_k(X) >_L \beta_k(Y)$, and therefore $P(X) >_L P(Y)$.

For E, H, and A the same argument applies: each block of the key is an order statistic, under $\geq_L$ or under $\geq$, of quantities that weakly increase, so every block weakly increases; the blocks together contain every entry of X, so some block changes, and the first that does strictly increases. □

3.3 Duality

This section shows that De Coninck and Van De Putte's four rules are not closed under the exchange of persons and states, and that adding P, the new rule, closes them. Let X^T denote the transpose of X; if X is $n \times m$, then X^T is $m \times n$. For a rule F, define its transpose F^T by $F^T(X) = F(X^T)$: it applies F with the roles of persons and states exchanged. Since every rule is defined for matrices of all sizes, F^T is well defined. A family of rules is *transpose-closed* if it contains the transpose of each of its members (up to equivalence of the induced rankings).

Proposition 1 (Duality) For every matrix X: (a) $E(X) = H(X^T)$; (b) $P(X) = A(X^T)$; (c) $B(X) = B(X^T)$. Hence $E^T = H$, $H^T = E$, $P^T = A$, $A^T = P$, and $B^T = B$.

Proof The rows of X^T are the columns of X, and its columns are the rows of X. Hence $r_s(X^T) = c_s(X)$ for each state s, and $c_i(X^T) = r_i(X)$ for each person i. (a)

$$H(X^T) = \mathrm{sort}_L(c_1(X^T), \dots, c_n(X^T)) = \mathrm{sort}_L(r_1(X), \dots, r_n(X)) = E(X).$$

(b) For $k = 1, \dots, m$, the k-th component of $A(X^T)$ is

$$\mathrm{sort}(c_1(X^T)[k], \dots, c_n(X^T)[k]) = \mathrm{sort}(r_1(X)[k], \dots, r_n(X)[k]),$$

the k-th component of $P(X)$.

(c) X and X^T contain the same entries. The second sentence follows because $(X^T)^T = X$. □

Proposition 2 (Distinctness) No two of B, E, H, A, and P induce the same ranking on every scenario.

Proof In De Coninck and Van De Putte's (2023, Fig. 6) scenario (Table 3), the five rules induce five different strict rankings (Table 4); the first four rows reproduce their own Table 1. □

Table 4 Rankings induced by the five rules

Rule	Ranking

Basic (*B*)	$b \succ d \succ c \succ a$
Holistic ex ante (*E*)	$d \succ a \succ b \succ c$
Atomistic ex post (*A*)	$d \succ b \succ a \succ c$
Holistic ex post (*H*)	$c \succ b \succ d \succ a$
Atomistic ex ante (*P*)	$b \succ c \succ d \succ a$

Corollary 1 (Minimal closure) (a) De Coninck and Van De Putte's four rules are not transpose-closed. (b) $\{B, E, H, A, P\}$ is the smallest transpose-closed family containing them.

Proof By Proposition 1, the transpose of A is P, which by Proposition 2 differs from each of the four; this gives (a). Any transpose-closed family containing the four therefore contains P. By Proposition 1, the five rules are transpose-closed. □

4 The structure of the family

4.1 Two rearrangements

This section shows that each atomistic rule is a holistic rule applied to a rearranged matrix. Let $X^{\downarrow}$ be the matrix obtained by sorting each column of X in nondecreasing order. In $X^{\downarrow}$, row k collects the k-th lowest outcome of every state, as if a single person occupied the k-th position in every state. Let $X^{\rightarrow}$ be the matrix obtained by sorting each row of X in nondecreasing order. In $X^{\rightarrow}$, column k collects every person's k-th worst outcome, as if everyone's k-th worst outcome occurred in the same state.

Two properties of matrices correspond to these operations.

Definition 6 (Fixed social ranking; common shocks) A matrix X has a fixed social ranking if the persons can be ordered as $i_1, \ldots, i_n$ so that, in every state s, the entry of person i_k is at most that of person i_{k+1} for every $k < n$; the worst-off person is then the same in every state. X has common shocks if the states can be ordered as $s_1, \ldots, s_m$ so that, for every person i, the entry in state s_k is at most that in state s_{k+1} for every $k < m$; a state that is bad for one person is then bad for everyone.

Both properties are instances of comonotonicity, a standard tool in decision theory, where it

grounds rank-dependent evaluation (Schmeidler, 1989).[8] $X^{\downarrow}$ always has a fixed social ranking, and $X^{\rightarrow}$ always has common shocks. In Table 1, a and c have both properties, and b has neither.

Observation 1 (Rearrangement) For every matrix X, $A(X) = E(X^{\downarrow})$ and $P(X) = H(X^{\rightarrow})$.

Proof Suppose first that X has a fixed social ranking, and call the k-th person in that order person (k). Then $c_s(X)[k]$ is the outcome of person (k) in state s, so α_k is person (k)'s ordered prospect. Since person (k) is weakly below person $(k+1)$ in every state, person (k)'s ordered prospect is coordinatewise below person $(k+1)$'s, so sort_L lists the prospects in the same order and $E(X) = A(X)$. Now $X^{\downarrow}$ has a fixed social ranking, and its columns are those of X rearranged within each state, so $A(X^{\downarrow}) = A(X)$. Therefore $A(X) = A(X^{\downarrow}) = E(X^{\downarrow})$.
The second identity follows by duality:

$$P(X) = A(X^T) = E((X^T)^{\downarrow}) = H(((X^T)^{\downarrow})^T) = H(X^{\rightarrow}). \quad \square$$

Corollary 2 (Factorization) For every matrix X: $E(X) = E(X^{\rightarrow})$, $A(X) = E(X^{\downarrow})$, $P(X) = H(X^{\rightarrow})$, and $H(X) = H(X^{\downarrow})$. Each non-basic rule is thus E or H applied after sorting each row or each column.

Proof Sorting each row of X leaves every $r_i(X)$ unchanged, and sorting each column leaves every $c_s(X)$ unchanged; this gives the first and last identities. The other two are Observation 1. □

Transposition is invertible: applying it twice returns the original matrix, so E and H determine each other, as do P and A. The rearrangements are not. All matrices that differ only in who receives which outcome within each state share the same $X^{\downarrow}$, so X, and with it $E(X)$, cannot be recovered from $X^{\downarrow}$ or from $A(X)$. What A discards is which person occupies each position across states; by Proposition 3 below, this loss affects the key if and only if the persons' ranking differs across states. Whether that information ought to matter is a normative question, taken up in Section 6.

4.2 Indifference as the least favorable assignment

This section shows what the atomistic rules lose by discarding identities. Let $\Pi(X)$ be the set of matrices obtained from X by permuting outcomes among persons within each state, and $\Sigma(X)$ the

[8] Two vectors are comonotonic if they never move in opposite directions: whenever one is strictly higher in coordinate i than in coordinate j, the other is not strictly lower there. A fixed social ranking means that the columns of X are pairwise comonotonic; common shocks means that the rows are.

set obtained by permuting each person's outcomes among states. These are De Coninck and Van De Putte's (2025) Π- and Σ-variants, applied to a single matrix rather than to a whole scenario. Matrices in $\Pi(X)$ have the same state distributions as X but assign them to persons differently; matrices in $\Sigma(X)$ give each person the same prospect but assign it to states differently.

Proposition 3 (Least favorable assignment) For every matrix X : (a) $A(X) = \min\{E(Y): Y \in \Pi(X)\}$, and $A(X) \leq_L E(X)$, with equality if and only if X has a fixed social ranking. (b) $P(X) = \min\{H(Y): Y \in \Sigma(X)\}$, and $P(X) \leq_L H(X)$, with equality if and only if X has common shocks.

Proof (a) *Inequality and equality condition.* By induction on n; the case $n = 1$ is trivial. Let f be the vector of column minima of X, and let $r_{(1)}$ be the lowest of the $r_i(X)$ under $\geq_L$, the first block of $E(X)$. Every row of X is weakly above f in every state, so every $r_i(X)$ is coordinatewise above $\mathrm{sort}(f) = \alpha_1$, and therefore $\alpha_1 \leq_L r_{(1)}$. If the inequality is strict, $A(X) <_L E(X)$, and X has no fixed social ranking, since otherwise $E(X) = A(X)$ by the proof of Observation 1. Otherwise, some person i has $r_i(X) = \mathrm{sort}(f)$ while being weakly above f in every state; since the two vectors have the same sum, person i's outcomes equal f, so i is weakly lowest in every state. Removing row i removes the first block of both keys, and the remaining matrix has a fixed social ranking if and only if X does. The induction hypothesis gives the inequality and the equality condition.

Minimum. Every $Y \in \Pi(X)$ has the same sorted columns as X, so $A(Y) = A(X) \leq_L E(Y)$, and $X^{\downarrow} \in \Pi(X)$ attains $E(X^{\downarrow}) = A(X)$ by Observation 1.

(b) follows from (a) by Proposition 1, since transposition maps Π to Σ and fixed social rankings to common shocks. □

The atomistic ex post rule is indifferent to who occupies each social position. It evaluates every policy as the holistic ex ante rule would under the least favorable assignment of persons to positions, namely the one in which each position is held by the same person in every state. Dually, the atomistic ex ante rule evaluates every policy as the holistic ex post rule would under the least favorable assignment of each person's outcomes to states. Following positions rather than persons, as Rawls does with representative persons, amounts to assuming the worst about who occupies them.

4.3 Two choices

This section shows that the four non-basic rules are generated by two independent choices, and argues that ex ante and ex post should name only one of them. By Corollary 2 and Table 5, each non-basic rule applies a nested leximin that compares rows first (E) or columns first (H), after sorting each row ($\rightarrow$) or each column ($\downarrow$). The first choice is the information the rule retains: either it tracks persons (E and P), or it tracks states (H and A). The second is the priority: whether the outermost lexical comparison begins with the worst-off person or position (E and A), or with the worst state or worst case (H and P). The information choice is the ex ante/ex post distinction in its informational sense, the reading adopted in Section 2.2 (Proposition 4 below).

The informational reading is preferable for three reasons. First, it classifies a rule by its ranking alone, whereas the procedural reading depends on how the rule is written. By Corollary 2, the atomistic ex post rule can be computed state by state, as De Coninck and Van De Putte define it, or row by row on a rearranged matrix, so the procedural reading would call it both ex post and ex ante. The informational reading concerns what the ranking depends on instead of the order of operations, and so calls it ex post however it is computed. Second, the standard objections on each side concern information: the ex ante objection to ex post rules is that they ignore who bears the risk, and the ex post objection to ex ante rules is that they ignore whether risks are shared. A reading that distinguishes the two by information therefore captures fully the disagreement. Third, the procedural reading attributes to the ex ante/ex post contrast disagreements in which information plays no role, which the informational reading assigns to priority instead, as Table 6 and Corollary 3 show next.

Table 5 The four non-basic rules (Table 2 with the factorization of Corollary 2)

	Tracks persons (ex ante)	**Tracks states (ex post)**
Priority to the worst-off	E: worst-off person, $E(X) = E(X^{\rightarrow})$	A: bottom position, $A(X) = E(X^{\downarrow})$
Priority to the worst case	P: worst cases, $P(X) = H(X^{\rightarrow})$	H: worst state, $H(X) = H(X^{\downarrow})$

Table 5 shows that the atomistic/holistic distinction records how the two choices relate. A rule is atomistic when its priority dimension is one whose identities it discards. The atomistic ex post rule gives priority to the worst-off but does not know which outcomes belong to the same person, so it can locate the worst-off only by rank. The atomistic ex ante rule gives priority to the worst case

but does not know which outcomes occur in the same state, so it can locate the worst case only by rank. A rule is holistic when its priority dimension is one whose identities it keeps, so it can locate the worst-off person or the worst state by identity rather than by rank. Transposition exchanges the two dimensions and therefore swaps both choices, mapping E to H and P to A; the basic rule, which retains neither linkage, is self-dual.

The two choices are independent: all four combinations occur, and Proposition 2 shows that they differ. Moreover, the priority choice operates even when the information choice makes no difference.

Table 6 isolates the priority choice. Two persons face two policies whose results depend on whether the economy turns out good or bad. The lowest outcome under both policies is person 1's 1 in the bad economy, so a lexical rule must decide what to compare next. If it compares the prospects of the worst-off person, person 1's (1, 3) against (1, 4), it favors Policy 2. If it compares the distributions in the worst state, (1, 3) against (1, 2), it favors Policy 1. Yet nothing in Table 6 depends on keeping track of who is who or which state is which: person 1 is never better off than person 2, and the bad economy is better for no one. The disagreement must therefore be about priority. Corollary 3 makes this precise.

Table 6 A two-person, two-state example (welfare of persons 1 and 2)

	Good economy	**Bad economy**
Policy 1	(3, 3)	(1, 3)
Policy 2	(4, 4)	(1, 2)

Corollary 3 (Independence of priority) Let C be a scenario. If every alternative has a fixed social ranking, then $S_E(C) = S_A(C)$; if every alternative has common shocks, then $S_H(C) = S_P(C)$. When both hold, $S_E(C)$ and $S_H(C)$ can still differ.

Proof Under a fixed social ranking, $A(X) = E(X)$ for every alternative by Proposition 3(a); under common shocks, $P(X) = H(X)$ by Proposition 3(b). Equal keys give equal choice sets. In Table 6, both policies have both properties, yet S_E selects Policy 2 and S_H selects Policy 1. □

In general, by Observation 1, $S_A(C) = S_E(C^{\downarrow})$ and $S_P(C) = S_H(C^{\rightarrow})$, where $C^{\downarrow}$ and $C^{\rightarrow}$ rearrange every alternative.

4.4 Diamond's case decomposed

Return to Diamond's case (Table 1), which De Coninck and Van De Putte (2023, Fig. 7) adapt from Diamond's (1967) two-policy example by adding a third alternative, following Ben-Porath et al. (1997; see also Mongin & Pivato, 2021).

The case tests two things. The first is the comparison between a and c: inequality without common risk against equality with common risk. Both alternatives have a fixed social ranking and common shocks, so information plays no role, and by Corollary 3 the comparison is decided solely by the priority choice. The rules that give priority to the worst-off prefer c, whose worst-off person has the prospect $(0,1)$ rather than $(0,0)$; the rules that give priority to the worst case prefer a, whose worst state is $(0,1)$ rather than $(0,0)$.

The second is the value of spreading the loss. Alternative b has neither property, and its two rearrangements are the other alternatives: $b^{\downarrow} = a$ and $b^{\rightarrow} = c$. The rules that retain person-linkage see that b gives each person the prospect $(0,1)$, as c does, and evaluate b as c. The rules that retain state-linkage see that b has the same state distributions as a, and evaluate b as a. The information choice therefore decides whether b is evaluated as a or as c; the priority choice then decides whether that helps b. Of the four rules, only E, which retains person-linkage and gives priority to the worst-off, values spreading the loss.

Table 7 Diamond's case: the two choices fix every verdict

Rule	***a* vs. *c***	***b* evaluated as**	**Ranking**	**Admissible**
Basic (*B*)	tie	—	$a \sim b \sim c$	*a*, *b*, *c*
Holistic ex ante (*E*)	*c*	*c*	$b \sim c \succ a$	*b*, *c*
Atomistic ex post (*A*)	*c*	*a*	$c \succ a \sim b$	*c*
Holistic ex post (*H*)	*a*	*a*	$a \sim b \succ c$	*a*, *b*
Atomistic ex ante (*P*)	*a*	*c*	$a \succ b \sim c$	*a*

Combining the two components yields every ranking and admissible set in Table 7, and these agree with those computed directly. The case usually serves as a test of ex ante against ex post evaluation, and its second component is indeed such a test. But the case also tests the priority choice, and that test separates rules that agree on the information choice. E and P both retain person-linkage and both evaluate b as c; yet E rejects the persistent loser a, whereas P selects it uniquely.

The decomposition also accounts for De Coninck and Van De Putte's own reading of the case. They read it through three intuitions (2023, p. 296): a is worse than b and c in terms of ex ante equality; b is worse than c in terms of ex post equality; and, by what they call ex post lexical optimality, c is worse than both a and b. Each intuition is the verdict of one of their rules: the first is the holistic ex ante rule's, the second the atomistic ex post rule's, and the third the holistic ex post rule's. In terms of Table 2, the tension they note between the last two is the priority choice: both rules are ex post and differ only in giving priority to the worst-off or to the worst case. No intuition they list matches the atomistic ex ante rule, which selects a.

They also suggest that a lexicographic combination, first ex ante welfare and then ex post equality, might yield the ranking $c \succ b \succ a$. Within the present family, it does in Diamond's case: the holistic ex ante rule followed by the atomistic ex post rule yields exactly that ranking. Both stages give priority to the worst-off and differ only in information, so the combination uses both linkages under a single priority. Combining the two holistic rules instead, in either order, selects b uniquely, because they differ in priority as well.

5 Information and the original position

De Coninck and Van De Putte (2025) show that the standard original position supports only rules that erase both kinds of linkage. This section restates their result in terms of the two choices of Section 4: among the rules studied here, the original position can make neither the information choice nor the priority choice.

Their argument runs as follows. Behind the veil, a party is uncertain both about the state and about which person they will be. De Coninck and Van De Putte combine the two in the *original-position transformation*, which replaces each state by every pairing of that state with an assignment of identities to persons; in each such pair, a party receives the welfare of whichever person they turn out to be. A correspondence S can be supported by an original-position argument if some individual choice rule meeting their conditions (Individualism, Symmetry, and State Replication Indifference) selects exactly $S(C)$ for a party facing the transformation of C, for every scenario C (De Coninck & Van De Putte, 2025, Definitions 8, 14, and 15). Each of the five correspondences is a social choice rule in their sense (2025, Definition 13). Its key depends only on sorted outcomes, so it ignores the labels of states and alternatives. Replicating every state the same number of times

repeats each entry of the key the same number of times, which leaves every lexicographic comparison unchanged.

They characterize these correspondences by two indifference conditions. A correspondence S satisfies IISD if $S(C) = S(C')$ whenever C' is obtained from C by permuting, in each state, outcomes among persons (the same permutation for every alternative). It satisfies IIPD if $S(C) = S(C')$ whenever C' is obtained by permuting, for each person, outcomes among states (the same permutation for every alternative). A correspondence that violates IISD uses person-linkage; one that violates IIPD uses state-linkage. In the terms of Section 2.2, IIPD is implied by the informational reading of ex ante evaluation, and IISD by that of ex post evaluation; De Coninck and Van De Putte themselves also read IISD as a form of ex post anonymity (2025, Section 3.3). They classify several rules, including three expected-utility rules, by these conditions (2025, Table 2); Proposition 4 does the same for the five lexical rules, which their paper does not classify.

> **Proposition 4** (Information) (a) S_E and S_P satisfy IIPD but violate IISD. (b) S_A and S_H satisfy IISD but violate IIPD. (c) S_B satisfies both.

Proof Permuting a person's outcomes across states leaves every $r_i(X)$ unchanged, so E and P are unchanged; permuting outcomes among persons within a state leaves every $c_s(X)$ unchanged, so A and H are unchanged; and both permutations leave the entries of X unchanged, which settles (c).

For the violations, swap the two persons' outcomes in state 2 of Diamond's case. This maps a to b, b to a, and c to itself, so S_E changes from $\{b, c\}$ to $\{a, c\}$ and S_P from $\{a\}$ to $\{b\}$. By Proposition 1, transposing this witness gives violations of IIPD for S_H and S_A. □

> **Corollary 4** Of the five correspondences, only S_B can be supported by an original-position argument.

Proof By Proposition 4, S_E and S_P violate IISD and S_A and S_H violate IIPD, while S_B satisfies both. The claim follows from their characterization (2025, Corollary 2): a correspondence can be supported if and only if it satisfies both conditions. □

For B, E, A, and H, the conclusion is De Coninck and Van De Putte's (2023, Section 4), proved there by a direct argument; the new case is P.

Since the only supported rule, B, makes neither choice, the original position supports none of the

rules here that make either. Making the information choice would require a model that keeps persons apart from states, which is what De Coninck and Van De Putte call for. However, even with such a multidimensional model, making the priority choice would require a further reason to put the worst-off before the worst case, or the reverse.

6 What the two choices commit a Rawlsian to

In Rawls's own setting, with identity uncertainty alone (Section 2.3), the worst-off and the worst case coincide at every lexical level. With world uncertainty they still agree on the lowest outcome, which is why the simple difference principle has a single specification (Section 2.4). But the following lexical comparison, as I have argued in Section 4.3, becomes ambiguous. Rawls's main argument in *A Theory of Justice* reaches the least advantaged through the maximin rule for choice under uncertainty (Rawls, 1999, Section 26), and his later reasons for the maximin equity criterion are independent of that rule (Rawls, 1974, p. 141). Neither says how to continue after the lowest outcome, or whether the least advantaged should be followed as a person across states or as a position. Rawls's reasoning thus settles neither choice.

A Rawlsian must therefore make them on further moral grounds. For instance, the atomistic ex post rule carries Rawls's practice of evaluating representative positions rather than persons over to world uncertainty. However, its cost is that it cannot value spreading a loss across persons. A Rawlsian who regards positions as the proper objects of concern should accept this consequence explicitly.

The atomistic ex ante rule has some appeal. It tracks persons, as ex ante fairness requires. However, its cost is that it uniquely selects the alternative in which the same person loses in every state in Diamond's case. Rejecting it while accepting its transpose, the atomistic ex post rule, is consistent only if persons and states differ in moral significance; since the two rules have the same form, that difference must be argued for.

Transposition of the person-by-state matrix may seem a trivial operation, but it is a systematic way of finding counterparts: here it finds the atomistic ex ante rule. Rawls objects that utilitarianism does not take seriously the distinction between persons (Rawls, 1999, p. 24), and one might think that treating persons and states symmetrically repeats the error. It does not. The duality is technical, saying that certain rules are the same operation applied along different dimensions, not that the

dimensions matter equally. On the present analysis, an argument that persons and states differ must be made twice, once for the information a principle uses and once for the priority it gives. The priority choice in particular has received little attention, although Corollary 3 shows that it can decide a case on its own.

7 Conclusion

Under uncertainty, the lexical difference principle splits into specifications that rank policies differently. This paper has shown that these specifications are generated by two choices: whether a rule tracks persons or states, and whether lexical priority goes to the worst-off or to the worst case. De Coninck and Van De Putte's non-basic specifications cover three of the four combinations; the fourth, the atomistic ex ante rule, completes the family and is found by exchanging the roles of persons and states. The holistic rules arise when the two choices concern the same dimension, and the atomistic rules when they diverge. Each atomistic rule is its holistic counterpart evaluated under the least favorable rearrangement of persons or of states.

Neither the standard original position nor Rawls's own reasoning settles either choice, so a Rawlsian who favors a particular lexical principle must defend both. Several questions remain open. First, the operations used here can be combined: sorting rows and then columns, for example, yields further rules that track persons but differ from both ex ante rules, and lexicographic combinations of rules retain both linkages (Section 4.4). Mapping this larger family, and choosing among its members on normative grounds, is left for future work. Second, outcomes may be indexed by further dimensions, such as time or kinds of primary goods; each can again be treated as an axis of the matrix, and transposition generalizes to permutations of the axes. Checking every permutation of the dimensions then ensures that the family considered is closed. Third, under unequal probabilities, transposition loses its symmetry, but the two choices still apply, so the structure could be studied in a probabilistic setting.

Statements and Declarations

Code availability The rankings and admissible sets reported in Section 1 and in Tables 4 and 7 were computed directly. Observation 1, Corollary 2, and the inequalities and equality conditions of Proposition 3 were additionally checked on every matrix of size 2×2 with welfare levels 0 to 3, and on every matrix of sizes 2×3, 3×2, 3×3, 2×4, and 4×2 with welfare levels 0 to 2. The

minimum in Proposition 3(a) was checked on random matrices of sizes 2×2, 2×3, and 3×2, and Propositions 1 and 4, Lemmas 1 and 2, and the claims of Section 4.4 were checked on randomly generated matrices with up to four persons and four states. No counterexample was found. The Python code is available from the author on request.

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